Quiz 4 solutions

Q1

1 Point

An additive model has **lower variance** than a linear model fit on the same data. (You may assume that the additive model is spline based with finite curvature penalty $\lambda > 0$ on each term to avoid silly solutions.)

- ○ TRUE
- ◉ FALSE

Additive models generally decrease bias by allowing non-linearity, but increase variance because of the added flexiblity.

Q2

1 Point

The model

$$\hat{f}(x_1, x_2) = 2 - 3x_1 + 5x_2 - 2x_2^2$$

is an example of an additive model in x_1, x_2.

- ◉ TRUE
- ○ FALSE

We can write the function as two pieces added together, the first of which is a function only of x_1 and the second only of x_2.

Q3

2 Points

Suppose that I build a classifier to guess whether the S&P 500 will increase by at least 1% on a given day. I carefully test my classifier on new data and discover that I only have 10% misclassification error! Does this mean that I have a good classifier? Explain.

Not necessarily. Several arguments are acceptable. (1) That we might not have symmetric losses for whatever our intended purpose is, so don't care about misclassification rate. (2) That we don't know the base rate of jumps of this size, and might do better just guessing (e.g.) that it never jumps.

Q4

3 Points

Suppose that we have the following discrete joint distribution over binary $X_1 \in \{0,1\}$, $X_2 \in \{0,1\}$, and $Y \in \{0,1\}$.

$\mathbb{P}(X_1 = 0, X_2 = 0, Y = 1) = 0.2$
$\mathbb{P}(X_1 = 1, X_2 = 0, Y = 1) = 0.002$
$\mathbb{P}(X_1 = 0, X_2 = 1, Y = 1) = 0.02$
$\mathbb{P}(X_1 = 1, X_2 = 1, Y = 1) = 0.25$
$\mathbb{P}(X_1 = 0, X_2 = 0, Y = 0) = 0.3$
$\mathbb{P}(X_1 = 1, X_2 = 0, Y = 0) = 0$
$\mathbb{P}(X_1 = 0, X_2 = 1, Y = 0) = 0.025$
$\mathbb{P}(X_1 = 1, X_2 = 1, Y = 0) = 0.203$

What is the Bayes classifier $f(X_1, X_2)$ that predicts the class $Y \in \{0,1\}$? Completely specify the function.

$f(X_1, X_2) = 1$ for which (X_1, X_2) pairs?

To predict 1, $P(X, Y = 1)$ must be bigger than $P(X, Y = 0)$. This occurs for (X_1, X_2) pairs: $(1,0), (1,1)$.

$f(X_1, X_2) = 0$ for which (X_1, X_2) pairs?

To predict 1, $P(X, Y = 1)$ must be smaller than $P(X, Y = 0)$. This occurs for (X_1, X_2) pairs: $(0,0), (0,1)$.

Q5

2 Points

Suppose that you have n data points (x_i, y_i), $x_i, y_i \in \mathbb{R}$. You decide to fit a smoothing spline, which solves:

$$\sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \int (f''(x))^2 \, dx.$$

As $\lambda \to \infty$, what does the solution converge to?

Least squares line

As $\lambda \to 0$, what does the solution converge to?

A function that interpolates the training point. You may also point out that the particular interpolating function is a cubic spline, but that is not necessary.